what can be computed?

with what kind of computer?

All comp probs can be grouped:

Decision:

answer T/F

given G, u, v, is there a path from u to v?

Optimization:

answer some opt value

Given G, u, v what is length of shortest path from u to v?

search:

answer: some object

give a shortest u-v path

come up w/ a computational prob in one type

share w/ a neighbor: they fill in other 2 types

① Given a list, sort it. S
how many swaps to sort it? O
is it sorted? D

② Given G_1, G_2, give G_3 s.t. for every node in G_1 it is connected to some node in G_2 and vice versa. S

Given language L, is $w \in L$?

set membership problem

Is p prime?

Let L be the set of all binary reps. of prime #s.

$p_2 \in L \implies p$ is prime

Given G, is there a cycle in G?

Fix any way of representing graphs as binary strings.

Let L be the set of all binary reps. of graphs w/ cycles.

$R(G) \in L \iff G$ has a cycle.

DoSomethingCool():
$q \leftarrow 0$

q	$\delta[q,0]$	$\delta[q,1]$	$A[q]$
0	0	1	True

Context-free Languages

A language is context-free if it can be generated by a context-free grammar.

ex

$S \to A$

$S \to B$

$A \to 0A$

$A \to 0C$

$B \to B1$

$B \to C1$

$C \to \varepsilon$

$C \to 0C1$

$L(C)$

S
A
$0A$
$00A$
$000A$
$0000A$
$00000C$
$000000C1$
$0000000\varepsilon1$
00000001

Σ : symbols / alphabet / terminals $\{0, 1\}$

Γ : non-terminals $\{S, A, B, C\}$

Gamma

R : production rules $D \to w$, where $D \in \Gamma$ and $w \in (\Sigma, \Gamma)^*$

omega

$C \to 0C1 \leftarrow \varepsilon \in \Sigma$

$\in \Gamma \quad \in \Sigma \quad \in \Gamma$

$S \in \Gamma$ Starting non-terminal S

$L(D)$: language of D = set of all strings generated by D

$L(C)$?

$\{0^n 1^n : n \geq 0\}$

C

ε

C

$0C1$

$00C11$

$S \to A \mid B$

$A \to 0A \mid 0C$

$B \to B1 \mid C1$

$C \to \varepsilon \mid 0C1$

$L(S)$?

```
START():
    if [redacted]
        ALPHA()
    else
        BRAVO()
```

```
ALPHA():
    if [redacted]
        print(0)
        ALPHA()
    else
        print(0)
        CHARLIE()
```

```
BRAVO():
    if [redacted]
        BRAVO()
        print(1)
    else
        CHARLIE()
        print(1)
```

```
CHARLIE():
    if [redacted]
        return
    else
        print(0)
        CHARLIE()
        print(1)
```